



USN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

AC2002-1 – DATA STRUCTURES AND ALGORITHMS

(For ACT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What happens when an array is passed as a parameter to a function in C?
A) A new copy of the array is created B) The address of the first element of the array is passed
C) The entire array is passed by value D) The array is automatically sorted
2. What would be the Prefix notation for the given equation? $A+(B*C)$
A) $+*ACB$ B) $+A*BC$
C) $*B+AC$ D) $*A+CB$
3. Suppose we have a one dimensional array, named 'x', which contains 10 integers. Which of the following is the correct way to allocate memory dynamically to the array 'x' using malloc()?
A) $x=(int*)malloc(10);$ B) $x=(int*)malloc(10,sizeof(int));$
C) $x=malloc(int 10,sizeof(int));$ D) $x=(int*)malloc(10*sizeof(int));$
4. A normal queue, if implemented using an array of size MAX_SIZE, gets full when?
A) $Rear = MAX_SIZE - 1$ B) $Front = (rear + 1) \bmod MAX_SIZE$
C) $Front = rear + 1$ D) $Rear = front$
5. Which of the following information is stored in a singly-linked list's nodes?
A) Value of node B) Address of next node
C) Value of node and Address of prev node D) Value of node and Address of next node
6. While evaluating a postfix expression, when an operator is encountered, what is the correct operation to be performed?
A) Push it directly onto the stack B) Pop 2 operands, evaluate them and push the result onto the stack
C) Pop the entire stack D) Ignore the operator
7. The prefix form of the infix expression $(A + B) * (C - D)$ is:
A) $AB+CD-*$ B) $*+AB-CD$
C) $+AB-CD*$ D) $*AB-CD+$
8. What is the order of insertion in a Binary Search Tree?
A) Arbitrary B) Increasing order
C) Depends on the implementation D) Depends on the value to be inserted
9. In an undirected graph with V vertices and E edges, what is the maximum number of edges possible?
A) $V*(V-1)/2$ B) V^*E
C) $E*(E-1)/2$ D) 2^*V
10. Which traversal algorithm is used in breadth-first search (BFS)?
A) Queue-based B) Stack-based
C) Linked List D) Recursion

11. Which traversal method is used to print the elements of a binary search tree in ascending order?
- A) Pre-order Traversal B) Post-order Traversal
C) In-order Traversal D) Level-order Traversal
12. Which of the following is a property of a minimum spanning tree?
- A) It contains no cycles B) It contains all the edges of the original graph
C) It always has the largest weights D) It has fewer vertices than the original graph
13. What does Big O notation describe?
- A) The best case complexity of an algorithm B) The average case complexity
C) The worst case complexity D) The total execution time
14. What is the time complexity of Breadth First Search (BFS) on a graph with V vertices and E edges?
- A) $O(V+E)$ B) $O(V^2)$
C) $O(E^2)$ D) $O(V \log E)$
15. Which of the following is an example of a divide and conquer algorithm?
- A) Binary Search B) Linear Search
C) Bubble Sort D) Insertion Sort
16. What is a characteristic of the greedy algorithm?
- A) It makes the locally optimal choice at each step B) It solves subproblems recursively
C) It guarantees an optimal solution in all cases D) It always has linear time complexity
17. Which of the following sorting algorithms has the best average case time complexity?
- A) Bubble Sort B) Insertion Sort
C) Merge Sort D) Quick Sort
18. In dynamic programming, how are subproblems solved?
- A) Recursively B) Iteratively
C) By storing results of solved subproblems D) By brute force
19. What kind of algorithm is Dijkstra's algorithm?
- A) Divide and Conquer B) Greedy Algorithm
C) Dynamic Programming D) Backtracking
20. Which of the following is solved using dynamic programming?
- A) Traveling Salesman Problem B) Quick Sort
C) Binary Search D) Prim's Algorithm

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Develop a C program to define a structure Student with members name, age, and marks. Dynamically allocate memory for a student using a structure pointer, initialize the details, and display them.	6	L*3	1	1,2
	b) Explain the queue data structures and write a C program to implement enqueue, dequeue and display operations on linear queue using arrays.	10	L3	2	1,2
2.	a) Give the example for a sparse matrix and show how to represent a sparse matrix using arrays. Write a C program to convert the given sparse matrix into reduced sparse matrix.	8	L3	1	1,2
	b) Explain stack data structure. Write a C program to implement push, pop and display operation on stack using arrays.	8	L3	2	1,2
3.	a) Demonstrate, with a C program, how to send an array as an argument to a function.	6	L2	1	1,2
	b) Develop a C program to perform insertion, deletion, display and search operation on a singly linked list.	10	L3	2	1,2

Unit – II

- | | | | | |
|--|----|----|---|-----|
| 4. a) Describe the Binary Search Tree (BST) data structure with an example. Write a C program to implement a Binary Search Tree (BST) by writing functions for createNode, insertion, and search operations. | 10 | L3 | 3 | 1,2 |
| b) Explain Big O, Big Omega and Big Theta asymptotic notations with appropriate examples. | 6 | L2 | 4 | 1,2 |
| 5. a) Explain the key terminologies of tree data structure with relevant diagram. | 6 | L2 | 3 | 1,2 |
| b) Demonstrate how to traverse a graph using Depth First Search (DFS) algorithm/Program with suitable example. | 10 | L3 | 4 | 1,2 |
| 6. a) Compare and contrast directed graph, undirected graph and weighted graph. Write the example for each along with the adjacency matrix. | 6 | L2 | 3 | 1,2 |
| b) For n=5, calculate the number of recursive calls made by a factorial function and analyze the time complexity. | 5 | L2 | 4 | 1,2 |
| c) Analyze the time complexity of the "for loop" for different scenarios. | 5 | L2 | 4 | 1,2 |

Unit – III

- | | | | | |
|---|---|----|---|-----|
| 7. a) Demonstrate, with an example, how to sort an array using the Merge Sort algorithm and analyze its time and space complexity. | 8 | L2 | 5 | 1,2 |
| b) Develop an algorithm/program to find the single source shortest path using greedy approach (Dijkstra's Algorithm) and explain with an example. | 8 | L3 | 5 | 1,2 |
| 8. a) Apply Binary search algorithm to search an element 75 in the below array and illustrate its time complexity.
60, 5, 75, 7, 8, 1, 23, 13, 6, 12 | 8 | L3 | 5 | 1,2 |
| b) Discuss the role of dynamic programming in solving i) longest common subsequence and ii) chain matrix multiplication, and explain how it is different from greedy solutions. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
